CRISP-DM framework:
Data Analysis Pipeline for Precision Nutrition Analysis of Early Childhood Growth Trajectories

NO	PROCESS	DESCRIPTION
1	**Project Understanding**	
	Objective	**Research Question**: Are maternal dietary and socioeconomic factors associated with early childhood growth based on BMI-for-age z scores (zBMI)? **Population:** Children aged 24 months (n = 300) in the study cohort with available growth and maternal characteristics. **Exposures**: Maternal dietary and socioeconomic factors (i.e., ultra-processed food, energy intake, and household income) **Outcome**: zBMI (24 months) clusters
	1a. What exactly is the problem, the expected benefits?	Early childhood represents a critical window for growth and metabolic programming, during which nutritional adequacy exerts influences on long-term health trajectories. Inadequate or excess nutrition during this period is associated with malnutrition or accelerated risks in metabolic diseases later in life. Identifying factors associated with early-life growth patterns could help identify children at risk of adverse health outcomes and inform the development of public health and clinical guidelines. Growth trajectory assessments is therefore an important analytic tool. However, the heterogeneity of early childhood growth patterns often limits the ability to detect meaningful subgroups, and a data-driven framework capable of capturing this heterogeneity is needed. Maternal factors, including dietary quality, energy intake, and socioeconomic status, have been shown to affect infant growth. However, the degree to which these factors shape the children's growth trajectories in the first 24 months of life remains incompletely understood. The proposed approach examines the maternal factors, including UPF consumption, total energy intake, and household income, as modifiable and measurable predictors of heterogeneous growth patterns. By grouping children according to similar growth patterns, our framework supports the targeted disease prevention strategies and promote healthy dietary practices during gestation and early life. **Expected benefits:** • Better understand how maternal diet and socioeconomic factors influence growth.

		• Identifying heterogeneous growth patterns and clustering zBMI scores will allow for the detection of subsets of children that may respond differently to our exposures (i.e., maternal dietary, socioeconomic, clinical factors, and gut microbiome) • Inform precision nutrition strategies and targeted early-life interventions rather than generalized population-based recommendations. This approach aligns with precision nutrition by identifying subgroups of children who may respond differently to environmental and maternal exposures.
	1b. How would a solution look like?	Children will be clustered by their zBMI scores at 24 months, and these clusters will be used to assess whether maternal dietary and socioeconomic factors are associated with zBMI. • A set of distinct child clusters will be compared against maternal dietary and socioeconomic profiles to determine if these maternal factors predict zBMI score at 24 months. • A visual summary (dendrogram) that clearly shows cluster differences in zBMI and associated exposures. • Interpretation for each cluster, e.g. Cluster 1 = median zBMI score, cluster 2 = high zBMI, and 3 = low zBMI From these analyses, we may be able to extract useful information regarding which factors strongly associate and their directionality, and to establish tailored intervention recommendations.
	1c. What is known about the domain?	Early-life growth patterns are known predictors of later obesity and metabolic disease risk, and maternal nutrition and socioeconomic conditions have been shown to influence early childhood growth and development. The zBMI scores being utilized in this dataset are based on the WHO Child Growth Standards is an internationally accepted tool for classifying nutritional status in children under 5.
2	**Data Understanding**	
	2a. What data do we have available?	**Maternal dietary factors:** • Energy intake (kcal) – continuous • Fiber intake (g) – continuous • Protein intake (g) – continuous • Ultra-processed score – continuous • Dietary pattern score – continuous **Microbiome factors:** • Firmicutes : Bacteroidetes ratio – continuous • Shannon diversity – continuous • SCFA index – continuous

		Socioeconomic factors: • Household income – continuous • Maternal education (years) – ordinal **Maternal health factors:** • Maternal BMI (kg/m^2) – continuous • Gestational age (weeks) – continuous **Clinical biomarkers:** • Fasting glucose (mmol/L) – continuous • C-reactive protein (CRP) (mg/L) – continuous • Alanine aminotransferase (ALT) (U/L) – continuous **Child growth variables:** • Age at 24 months (months) – categorical • Sex – categorical • Weight at 0 (birth), 6, 12, and 24 months (kg) – continuous • Length at 0 (birth), 6, 12, and 24 months (cm) – continuous • BMI & BMI-for-age z-scores (zBMI) at 0, 12, and 24 months – continuous • Stunted status at 24 months – categorical
	2b. Is the data relevant to the problem?	Yes, the provided dataset includes 30 variables representing maternal dietary factors, socioeconomic factors, clinical biomarkers, and gut microbiome, which are possible determinants of growth and metabolic health. Maternal dietary and socioeconomic factors are being tested for their association with the outcome of interest (zBMI at 24 months).
	2c. Is it valid? Does it reflect our expectations?	The data leveraged is commonly used for clinical and nutritional analyses. However, several potential data quality issues exist, including: • Missing data • Implausible age values • Outliers in zBMI and clinical biomarkers • Data entry error The data leveraged for this analysis was evaluated using exploratory analyses (histograms and boxplots) to assess the distribution of variables, detect outliers, and evaluate if the values reflect realistic biological values. The missingness pattern was assessed to guide imputation.
	2d. Is the data quality, quantity, recency sufficient?	**Quantity:** The dataset contains 300 observations across 30 variables, providing a reasonable sample for exploratory clustering. However, there is a possibility of clusters with small numbers (e.g. *n* < 30) which are underpowered for biological interpretation.

		Quality: The dataset appears sufficient for exploratory growth analysis, although there are intentional imperfections (e.g., missing data, implausible values, biological outliers) that were assessed and addressed in the statistical model to improve the reliability of the data. **Recency:** Not applicable (mock precision nutrition cohort)
3	**Data Preparation**	
	3a. Which data should we concentrate on?	**Maternal dietary factors:** • Energy intake (kcal) • Ultra-processed score **Socioeconomic factors:** • Household income **Child growth variables:** • BMI-for-age z-scores (zBMI) at 24 months
	3b. How is the data best transformed for modelling?	The data will undergo several transformations, including: **Handling skewed data** • For any non-normally distributed variables, appropriate transformations will be applied (e.g., log-transforming data) to try to approximate normality to improve model stability • All relevant variables were assessed for normality using histograms and Shapiro-Wilk test. **Handling categorical data** • Categorical data was already transformed into a usable binary system for modelling. **Transforming BMI data** • BMI data does not require transformation as we are utilizing zBMI scores, a model standardized by the WHO using age and sex. Continuous variables included in clustering will be scaled prior to analysis to ensure comparability across variables with different measurement scales.
	3c. How may we increase the data quality?	Data quality can be improved through: • Identifying and evaluating statistical outliers **Growth data cleaning** • Removing implausible age values (e.g., negative values) • Applying WHO growth cut-offs to identify biologically implausible zBMI values (values that fall outside of WHO recommendations, -5 ≤ zBMI ≤ 5, will be flagged and removed from the dataset) (PMID: 15069916).

		Missing data • Data will be carefully assessed to determine if the values are missing completely at random, missing at random, or missing not at random • Any missing values will be assessed to determine if it is more appropriate to remove the entire record (deletion) or replace missing values using estimates (imputation) either by using mean/median imputation (replacing the missing values with the mean of the column), mode imputation (replacing the missing value with the most common, i.e., mode, value of the column) or using K-Nearest Neighbors (using KNN algorithm to find similar records and using their values to fill in missing data) → For the variables of interest, there were no missing data, so no imputations or KNN were applied. • If too many values are missing from a column, the variable may be omitted from analysis entirely à No values in the dataset were removed for missingness. Only biological outliers were removed as described earlier. Specifically, 4 individuals were removed for implausible age values, and 1 was removed for an implausible zBMI score.
4	**Modelling**	
	4a. What kind of model architecture suits the problem best?	zBMI at 24 months will be used to identify groups of children (i.e., clusters) with similar zBMI scores at 24 months of age. An unsupervised clustering approach was determined to be the most appropriate as the goal is to identify naturally occurring subgroups of children based on their zBMI scores and associated variables. Distance metrics (e.g., Euclidean distance) were used to measure similarity between individuals. All variables included in clustering were standardized (e.g., z-score scaling) to ensure equal contribution to distance calculations.
	4b. What is the best technique/ method to get the model?	Clustering approaches such as k-means or hierarchical clustering were used to identify distinct zBMI-based clusters. • K – Means: assign each child to the nearest centroid. • Hierarchical clustering: build a dendrogram to explore natural groupings without a fixed k. The ideal number of clusters will be further confirmed using statistical methods, specifically the elbow method and silhouette score.

		Multinomial logistic regression will be used to evaluate associations between maternal factors and cluster membership.
	4c. How good does the model perform technically?	Model performance will be evaluated using cluster validation metrics such as silhouette scores and examination of regression model assumptions (dependent variable is nominal (unordered) with 3+ categories, observations are independent, multicollinearity is imperfect, linearity of logit).
5	**Evaluation**	
	5a. How good is the model in terms of project requirements?	The model will be evaluated based on its ability to identify biologically meaningful growth patterns and whether maternal factors significantly explain differences in zBMI groups. Biological plausibility will be assessed by determining whether the identified clusters align with known profiles of undernutrition, normal growth, or overweight risk. **Evaluation of the clustering approach will be assessed by:** • Cluster tendency analysis (Hopkins statistic and PCA) • Silhouette score • Within-cluster variance • Visualization of zBMI score patterns **Evaluation of the regression models will be assessed by:** • R^2 • Residual diagnostics • Statistical significance of predictors (i.e., maternal dietary and socioeconomic factors)
	5b. What have we learned from the project?	The analysis will provide insight into how maternal diet and socioeconomic factors may influence early childhood growth patterns. Identifying these associations may help inform targeted early-life nutrition interventions. Precision nutrition insights from integrating multiple data domains (diet and socioeconomic) rather than single variables alone. Reproducibility is essential for transparent science and further research. These findings highlight the importance of considering inter-individual variability when developing early-life nutrition strategies.
6	**Deployment**	

	6a. How is the model best deployed?	The model could be used as a screening tool to identify maternal dietary and socioeconomic factors associated with unfavourable child growth patterns. Moreover, this model could be expanded to test other predictors (e.g., clinical biomarkers, gut microbiome factors) to aid in determining which maternal factors best predict zBMI at 24 months. In clinical or public health settings, such information could help guide targeted nutritional counselling during pregnancy and early childhood to support healthy growth patterns. This may support a shift in population-level recommendations toward subgroup-specific, personalized nutrition strategies. Ethical considerations for its implementation will also be made, particularly if any socioeconomic factors are found to be strongly associated with zBMI scores. Care must be taken to avoid stigmatization or inappropriate use of socioeconomic predictors in clinical decision-making. The reproducible R pipeline in GitHub ensures that any clinical site can re-run the same pipeline.
	6b. How do we know that the model is still valid?	Model validity would require testing the model on a new dataset to determine if similar growth patterns and associations are observed in a different population (i.e., using a test dataset that is independent of the training dataset). However, some limitations will exist, including potential residual confounding, measurement error in dietary reporting, and limited generalizability beyond the study cohort. Data pipeline integrity will ensure that any changes to WHO cut-off parameters or imputation technique are tracked, reproducible, and reviewable by collaborators and colleagues.
7	**What is your goal?**	The goal of this data pipeline is to identify distinct growth patterns in children at 24 months using BMI-for-age z-scores to evaluate whether maternal dietary and socioeconomic factors are associated with these patterns. By applying unsupervised clustering, the aim of this analysis is to uncover biologically meaningful subgroups and assess how maternal factors associate with early-life growth patterns. This approach consequently supports precision nutrition by enabling targeted, subgroup-specific intervention strategies.
8	**Which is the population/cohort that you are using?**	The study population consists of 300 children at 24 months of age from a mock precision nutrition cohort with data collected on maternal dietary and socioeconomic factors.

9	**Data structure/properties:**	The dataset comprises approximately 300 observations and 30 variables, including continuous, ordinal, and categorical data. The data requires preprocessing steps, including data transformation and scaling. zBMI is already standardized based on WHO growth references, allowing for direct comparison between individuals. The dataset exhibits moderate variability and mild skewness, with outliers identified and appropriately addressed during data cleaning.
10	**Gold Standard:**	While there is no gold standard for clustering-based identification of growth patterns, we use the WHO Child Growth Standards, which provides a clinically acceptable reference for interpreting growth during early childhood. Therefore, clustering validity is assessed based on alignment with clinically meaningful growth categories and internal validation metrics.